

SCHEDULER PRACTICAL

Node - Selector

1. Created a replica set without mentioning the node selector values. Pods were created in different nodes.

```
[ec2-user@ip-172-31-38-185 ~]$ sudo -i
[root@ip-172-31-38-185 ~]# kubectl get namespace
NAME          STATUS    AGE
default       Active    21m
kube-node-lease Active    21m
kube-public   Active    21m
kube-system   Active    21m
[root@ip-172-31-38-185 ~]# ls
[root@ip-172-31-38-185 ~]# vi file.yaml
[root@ip-172-31-38-185 ~]# kubectl apply -f file.yaml
replicaset.apps/my-replicaset created
[root@ip-172-31-38-185 ~]# kubectl get pod
NAME          READY   STATUS    RESTARTS   AGE
my-replicaset-d2k9l 1/1     Running   0           9s
my-replicaset-fpgnn 1/1     Running   0           9s
my-replicaset-vrxj6 1/1     Running   0           9s
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP            NODE                                NOMINATED NODE   READINESS GATES
my-replicaset-d2k9l 1/1     Running   0           20s   100.96.2.155   i-0aaa844a5dd3cb7fb               <none>            <none>
my-replicaset-fpgnn 1/1     Running   0           20s   100.96.3.1    i-06a1ff1f884705ffc               <none>            <none>
my-replicaset-vrxj6 1/1     Running   0           20s   100.96.1.176   i-01943d4266b859f58               <none>            <none>
[root@ip-172-31-38-185 ~]#
```

2. We try the **nodeselector** , First we should set the label in the node then we mention it in the yaml file.

```
NAME          READY   STATUS    RESTARTS   AGE   IP            NODE                                NOMINATED NODE   READINESS GATES
my-replicaset-d2k9l 1/1     Running   0           20s   100.96.2.155   i-0aaa844a5dd3cb7fb               <none>            <none>
my-replicaset-fpgnn 1/1     Running   0           20s   100.96.3.1    i-06a1ff1f884705ffc               <none>            <none>
my-replicaset-vrxj6 1/1     Running   0           20s   100.96.1.176   i-01943d4266b859f58               <none>            <none>
[root@ip-172-31-38-185 ~]# kubectl label nodes i-0aaa844a5dd3cb7fb name=rauf
node/i-0aaa844a5dd3cb7fb labeled
[root@ip-172-31-38-185 ~]# kubectl label nodes i-06a1ff1f884705ffc name=suhail
node/i-06a1ff1f884705ffc labeled
[root@ip-172-31-38-185 ~]# kubectl label nodes i-01943d4266b859f58 name=manoj
node/i-01943d4266b859f58 labeled
[root@ip-172-31-38-185 ~]#
```

3. We mentioned the Nodeselector in the yaml file while creating the 2nd replica set . It matches the node selector label and launches all the pods in the mentioned node.

```
root@ip-172-31-38-185:~
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: my-replicaset
spec:
  replicas: 3
  selector:
    matchLabels:
      app: myapp
  template:
    metadata:
      labels:
        app: myapp
    spec:
      nodeSelector:
        name: rauf # <-- Pod will run only on nodes with this label
      containers:
        - name: nginx
          image: nginx:latest

"file.yaml" 20L, 362B 4,22 All
```

Result :

```
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP            NODE                NOMINATED NODE   READINESS GATES
my-replicaset-d2k9l 1/1     Running   0           19m   100.96.2.155  i-0aaa844a5dd3cb7fb <none>           <none>
my-replicaset-fpgnn 1/1     Running   0           19m   100.96.3.1    i-06a1ff1f884705ffc <none>           <none>
my-replicaset-vrxj6 1/1     Running   0           19m   100.96.1.176  i-01943d4266b859f58 <none>           <none>
[root@ip-172-31-38-185 ~]# vi file.yaml
[root@ip-172-31-38-185 ~]# kubectl apply -f file.yaml
replicaset.apps/my-replicaset1 created
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP            NODE                NOMINATED NODE   READINESS GATES
my-replicaset-d2k9l 1/1     Running   0           20m   100.96.2.155  i-0aaa844a5dd3cb7fb <none>           <none>
my-replicaset-fpgnn 1/1     Running   0           20m   100.96.3.1    i-06a1ff1f884705ffc <none>           <none>
my-replicaset-vrxj6 1/1     Running   0           20m   100.96.1.176  i-01943d4266b859f58 <none>           <none>
my-replicaset1-g7dfc 1/1     Running   0           5s    100.96.2.53   i-0aaa844a5dd3cb7fb <none>           <none>
my-replicaset1-j9dqj 1/1     Running   0           5s    100.96.2.156  i-0aaa844a5dd3cb7fb <none>           <none>
my-replicaset1-xws7b 1/1     Running   0           5s    100.96.2.201  i-0aaa844a5dd3cb7fb <none>           <none>
```

Affinity : Node Affinity

4. Created a replica set without mentioning the node affinity values. Pods were created in different nodes.

```
[root@ip-172-31-38-185 ~]# kubectl get pod
No resources found in default namespace.
[root@ip-172-31-38-185 ~]# vi file.yaml
[root@ip-172-31-38-185 ~]# kubectl apply -f file.yaml
replicaset.apps/my-replicaset created
[root@ip-172-31-38-185 ~]# kubectl get pod
NAME                READY   STATUS    RESTARTS   AGE
my-replicaset-mwfgs  1/1     Running   0           10s
my-replicaset-nnwwl  1/1     Running   0           10s
my-replicaset-s49vk  1/1     Running   0           10s
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP              NODE                NOMINATED NODE   READINESS GATES
my-replicaset-mwfgs  1/1     Running   0           16s   100.96.3.180    i-06a1ff1f884705ffc <none>            <none>
my-replicaset-nnwwl  1/1     Running   0           16s   100.96.2.181    i-0aaa844a5dd3cb7fb <none>            <none>
my-replicaset-s49vk  1/1     Running   0           16s   100.96.1.204    i-01943d4266b859f58 <none>            <none>
```

5. Adding node affinity in the yaml file mentioned node label name =suhail (i-06a1ff1f884705ffc)

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: my-replicaset1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: myapp
  template:
    metadata:
      labels:
        app: myapp
    spec:
      affinity:
        nodeAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            nodeSelectorTerms:
              - matchExpressions:
                  - key: name
                    operator: In
                    values:
                      - suhail
      containers:
        - name: nginx
          image: nginx:latest
```

Result :

```
[root@ip-172-31-38-185 ~]# vi file.yaml
[root@ip-172-31-38-185 ~]# kubectl apply -f file.yaml
replicaset.apps/my-replicaset1 created
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP              NODE                NOMINATED NODE   READINESS GATES
my-replicaset-mwfgs  1/1     Running   0           7m50s   100.96.3.180    i-06a1ff1f884705ffc <none>            <none>
my-replicaset-nnwwl  1/1     Running   0           7m50s   100.96.2.181    i-0aaa844a5dd3cb7fb <none>            <none>
my-replicaset-s49vk  1/1     Running   0           7m50s   100.96.1.204    i-01943d4266b859f58 <none>            <none>
my-replicaset1-2rvjh 1/1     Running   0           22s     100.96.3.218    i-06a1ff1f884705ffc <none>            <none>
my-replicaset1-449mq 1/1     Running   0           22s     100.96.3.165    i-06a1ff1f884705ffc <none>            <none>
my-replicaset1-zqvdc 1/1     Running   0           22s     100.96.3.79     i-06a1ff1f884705ffc <none>            <none>
```

6. Here I mentioned the **wrong key value label in the yaml file but the Node affinity rule is Hard.**
(requiredDuringSchedulingIgnoredDuringExecution). See what happens!!

```
NAME                READY   STATUS    RESTARTS   AGE   IP        NODE   NOMINATED NODE   READINESS GATES
my-replicaset1-wvlq7 0/1     Pending   0           23s   <none>    <none> <none>           <none>
my-replicaset1-z7zq6 0/1     Pending   0           23s   <none>    <none> <none>           <none>
my-replicaset1-z152g 0/1     Pending   0           23s   <none>    <none> <none>           <none>
[root@ip-172-31-38-185 ~]# vi file.yaml
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP        NODE   NOMINATED NODE   READINESS GATES
my-replicaset1-wvlq7 0/1     Pending   0           3m21s <none>    <none> <none>           <none>
my-replicaset1-z7zq6 0/1     Pending   0           3m21s <none>    <none> <none>           <none>
my-replicaset1-z152g 0/1     Pending   0           3m21s <none>    <none> <none>           <none>
[root@ip-172-31-38-185 ~]#
```

Affinity : Pod Affinity

Cluster is Ready. I have one master node and three worker node.

INSTANCE GROUPS

NAME	ROLE	MACHINETYPE	MIN	MAX	SUBNETS
control-plane-us-east-2b	ControlPlane	t3.medium	1	1	us-east-2b
nodes-us-east-2b	Node	t3.medium	3	3	us-east-2b

NODE STATUS

NAME	ROLE	READY
i-02e1fee455a84f2c0	control-plane	True
i-0631061aa0b38d67c	node	True
i-091e4da67e244a904	node	True
i-0f17741c54fbbec1f	node	True

Your cluster rauf.k8s.local is ready

1. Launch one nginx pod as a frontend, so I set the label as **app : frontend**

```
root@ip-172-31-38-185:~
```

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  labels:
    app: frontend
spec:
  containers:
  - name: nginx
    image: nginx:latest
```

```
~
~
~
~
~
```

```
[root@ip-172-31-38-185 ~]# kubectl apply -f file.yaml
```

```
pod/nginx-pod created
```

```
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx-pod	1/1	Running	0	20s	100.96.2.92	i-091e4da67e244a904	<none>	<none>

```
[root@ip-172-31-38-185 ~]# kubectl get pods --show-labels
```

NAME	READY	STATUS	RESTARTS	AGE	LABELS
nginx-pod	1/1	Running	0	111s	app=frontend

2. I go to launch the backend pod, I need this pod also to launch in the same workernode. So i mentioned the previous pod label in the labelselector key value with In operator.

```
root@ip-172-31-38-185:~
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backendpod
  labels:
    app: backend
spec:
  affinity:
    podAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
      - labelSelector:
          matchExpressions:
          - key: app
            operator: In
            values:
            - frontend
        topologyKey: "kubernetes.io/hostname"
  containers:
  - name: nginx
    image: nginx:latest
```

```
~
~
```

Result: Both the pods were launched in the same worker node.

```
pod/backendpod created
[root@ip-172-31-38-185 ~]# kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
backendpod    1/1     Running   0          7s    app=backend
nginx-pod     1/1     Running   0          4m28s app=frontend
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP           NODE                                     NOMINATED NODE   READINESS GATES
backendpod    1/1     Running   0          17s   100.96.2.186 i-091e4da67e244a904 <none>           <none>
nginx-pod     1/1     Running   0          4m38s 100.96.2.92  i-091e4da67e244a904 <none>           <none>
```

Affinity : Pod Anti-Affinity

Anti pod-affinity is the opposite version of the pod affinity, it will separate the pods. Don't be launched in the same worker node.

1. I have a nginx pod with the label app=frontend.

```
root@ip-172-31-38-185:~
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP           NODE                                     NOMINATED NODE   READINESS GATES
nginx-pod     1/1     Running   0          19m   100.96.2.92  i-091e4da67e244a904 <none>           <none>
[root@ip-172-31-38-185 ~]# kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
nginx-pod     1/1     Running   0          19m   app=frontend
[root@ip-172-31-38-185 ~]#
```

2. Now i go to launch the backend pod with label app=frontend in the labelselector key value.

```
root@ip-172-31-38-185:~
apiVersion: v1
kind: Pod
metadata:
  name: backendpod
  labels:
    app: backend
spec:
  affinity:
    podAntiAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        - labelSelector:
            matchExpressions:
              - key: app
                operator: In
                values:
                  - frontend
            topologyKey: "kubernetes.io/hostname"
  containers:
    - name: nginx
      image: nginx:latest
~
~
~
```

3. Once I launch the backend pod, it doesn't launch in the same worker node (already the frontend pod launched node).

Result 👍: Both the pods launched in diff worker nodes.

```
[root@ip-172-31-38-185 ~]# kubectl apply -f file.yaml
pod/backendpod created
[root@ip-172-31-38-185 ~]# kubectl get pod -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
backendpod	1/1	Running	0	7s	100.96.3.248	i-0f17741c54fbbec1f	<none>	<none>
nginx-pod	1/1	Running	0	24m	100.96.2.92	i-091e4da67e244a904	<none>	<none>

```
[root@ip-172-31-38-185 ~]#
```